

Deep Learning based End-to-End Nonlinear Space-Time Adaptive Processing

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Abstract—Existing deep learning space-time adaptive processing (STAP) methods typically employ neural networks only to approximate high resolution space-time spectrum, while relying on traditional linear processing for the remaining stages. This hybrid design still suffers from performance degradation caused by model mismatch. To address this issue, we propose an end-to-end deep learning STAP framework that directly maps radar echoes to target detection statistics. Specifically, two subnetworks are first used to perform nonlinear pulse compression and space-time spectrum estimation, followed by a dual-branch subnetwork that produces the detection statistic for the cell under test. During training, a polynomial loss function is adopted to preserve global amplitude fidelity and critical target features. Simulation results demonstrate that the proposed method outperforms conventional nonlinear STAP techniques in clutter suppression and target detection, while significantly reducing computational complexity compared with sparse recovery approaches.

Index Terms—Space-time adaptive processing (STAP), deep learning, clutter suppression, target detection.

I. INTRODUCTION

SPACE-TIME adaptive processing (STAP) is a critical technology for clutter suppression in airborne early warning radar [1]–[3], and its performance hinges on the accurate estimation of the clutter covariance matrix (CCM). However, in practice, the true CCM is not directly obtainable. It is typically estimated using a substantial number of independent and identically distributed training samples via maximum likelihood estimation [4]. This process, essentially a linear averaging of the quadratic terms of the sample data, constitutes a linear processing technique.

In real clutter environments, the scarcity of homogeneous training samples often results in significant degradation of STAP performance. Reduced-dimension and reduced-rank STAP techniques address this issue by transforming the full degrees of freedom problem into a local degrees of freedom framework through linear transformations, thus alleviating demands on sample size and computational complexity [5]–[9]. Nonetheless, these methods still require considerable samples. Alternatively, knowledge assisted STAP techniques reduce sample dependency by incorporating weighted linear fusion of prior information with the estimated CCM [10]–[12]. However, the effectiveness of this approach is largely contingent upon the accuracy of the prior information.

In recent years, researchers have proposed various STAP methods based on nonlinear estimation [13]–[21]. Notably, sparse recovery STAP models the space-time spectrum super resolution problem as an inverse problem. By employing sparse recovery algorithms, it can achieve high resolution clutter space-time spectrum coefficients and perform effectively even with limited training samples. However, these methods tend to have low efficiency in CCM estimation due to their reliance on multiple iterative computations, and their accuracy is constrained by the predefined observation models. Conversely, deep learning STAP methods utilize neural networks to capture the complex nonlinear relationships between observed data and the super resolved space-time spectrum, enabling rapid and high precision CCM reconstruction under limited sample conditions [22]–[26]. Nevertheless, these approaches face challenges such as poor model interpretability and difficult training processes. The above methods either perform nonlinear estimation through sparse recovery or use deep learning for nonlinear fitting of the space-time spectrum coefficients, followed by linear reconstruction of the CCM using a predefined discrete space-time dictionary. However, in practical scenarios where array amplitude and phase errors or grid mismatches exist, such linear reconstruction often leads to model mismatches, resulting in a loss of STAP performance.

To address these issues, this letter proposes an end-to-end fully nonlinear STAP method based on a cascaded neural network framework, which offers the following advantages:

- 1) It achieves the nonlinear fitting from radar echo data to detection statistics, by utilizing a cascaded network to perform pulse compression, space-time spectrum estimation, and detection statistic generation following clutter suppression. This significantly reduces the performance loss caused by model mismatch in existing nonlinear STAP methods.
- 2) It avoids iterative computations, significantly reducing computational complexity compared to sparse recovery STAP methods.

II. SIGNAL MODEL

As shown in Fig. 1, consider a moving platform equipped with a radar flying at a constant altitude H with a speed v . The radar employs a uniform linear array with an inter element

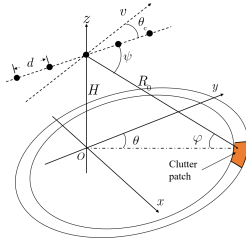


Fig. 1. Airborne radar viewing geometry.

spacing of d , transmitting waves with a wavelength of λ and a pulse repetition frequency of f_r . Let the yaw angle of the moving platform be θ_c , and the slant range to the clutter patch be R_0 . The corresponding azimuth angle, elevation angle, and spatial cone angle of the beam direction are denoted as θ , φ , and ψ , respectively.

Assuming the radar echoes encompass N array elements, K coherent pulses, and L range gates, the pre-compression radar echoes can be represented as a three dimensional data block $\mathbf{Y} \in \mathbb{C}^{N \times K \times L}$. If the pulse compression window length is P , the pulse compression process can be expressed as $x(n, k, l) = \sum_{p=-P}^{P-1} y(n, k, l)h(l-p)$, where $h(\cdot)$ denotes the pulse compression function. After pulse compression, the two dimensional slice of the radar data block for any range cell is rearranged into a column vector $\mathbf{x} \in \mathbb{C}^{N \times K \times 1}$, referred to as a space-time snapshot.

Typically, this snapshot is expressed as $\mathbf{x} = \mathbf{x}_t + \mathbf{x}_c + \mathbf{n}$, where $\mathbf{x}_t$, $\mathbf{x}_c$, and $\mathbf{n}$ represent the signal component, clutter component, and Gaussian white noise, respectively. Ignoring range ambiguity and uniformly dividing the range bin into N_c patches, the clutter component can be expressed as $\mathbf{x}_c = \sum_{i=1}^{N_c} \beta_i \mathbf{s}(\psi_i, f_{D,i})$, where β_i , ψ_i and $f_{D,i}$ are the complex amplitude, spatial cone angle, and normalized Doppler frequency of the i -th clutter patch, respectively. $\mathbf{s}(\psi_i, f_{D,i}) = \mathbf{s}_{t,i} \otimes \mathbf{s}_{s,i}$ is the space-time steering vector, where $\otimes$ denotes the Kronecker product, and

$$\mathbf{s}_{t,i} = [1, e^{j2\pi f_{D,i}}, \dots, e^{j2\pi(K-1)f_{D,i}}]^T \in \mathbb{C}^{K \times 1} \quad (1a)$$

$$\mathbf{s}_{s,i} = [1, e^{j2\pi f_{s,i}}, \dots, e^{j2\pi(N-1)f_{s,i}}]^T \in \mathbb{C}^{N \times 1} \quad (1b)$$

are the temporal and spatial steering vectors, respectively. Then, the CCM can be formulated by $\mathbf{R}_c = \mathbb{E}[\mathbf{x}_c \mathbf{x}_c^H]$, where $(\cdot)^H$ denotes the conjugate transpose. Assuming the CCM is known, the optimal STAP weight vector can be expressed as

$$\mathbf{w} = \frac{\mathbf{R}^{-1} \mathbf{s}(f_{s,t}, f_{D,t})}{\mathbf{s}(f_{s,t}, f_{D,t})^H \mathbf{R}^{-1} \mathbf{s}(f_{s,t}, f_{D,t})} \quad (2)$$

where $(\cdot)^{-1}$ is the matrix inversion, $\mathbf{R} = \mathbf{R}_c + \sigma^2 \mathbf{I}_{NK}$ is the CCM with diagonal loading considered, σ^2 is the noise variance, and $\mathbf{I}_{NK}$ is the NK dimensional identity matrix. Thus, the data from any Doppler channel after STAP processing can be represented as $\mathbf{z} = \mathbf{w}^H \mathbf{x}$.

III. DEEP LEARNING BASED NONLINEAR STAP

The nonlinear STAP signal processing workflow can be decomposed into three core components: nonlinear pulse com-

pression, nonlinear clutter space-time spectrum estimation, and the generation of nonlinear detection statistics. In previous studies, we have successfully implemented nonlinear pulse compression [27] and nonlinear clutter space-time spectrum estimation [26] for airborne early warning radar using a modular convolutional neural network (CNN). This letter focuses on the generation of nonlinear detection statistics based on deep learning and introduces a comprehensive end-to-end nonlinear STAP processing framework.

A. Nonlinear STAP Framework

The end-to-end nonlinear STAP framework is illustrated in Fig. 2. The framework takes the raw radar echo data cube (as shown in Fig. 2(a)) as input and directly outputs the detection statistic z for the cell under test (CUT). Specifically, the input data first undergoes nonlinear pulse compression along the fast-time dimension using the pulse compression network. Subsequently, the compressed data is segmented along the range dimension, and for each target cell, a processing data segment as shown in Fig. 2(b) is extracted, containing the clutter estimation data, guard cells, and target cell. Finally, the processing data segment is fed into the pipeline shown in Fig. 2(c), and the detailed processing flow is described as follows. The clutter estimation data is input to the spectrum estimation network to obtain the nonlinear clutter space-time spectrum estimate. This spectral information, along with the target cell data, is then jointly input into the detection statistics generation network, which ultimately synthesizes the detection statistic z . This completes the full end-to-end nonlinear STAP processing chain, from raw echoes to the detection statistic.

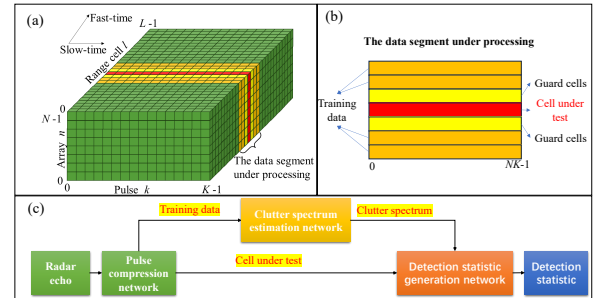


Fig. 2. Nonlinear STAP processing framework. (a) is the radar data cube. (b) is the data segment under processing. (c) is the end-to-end nonlinear STAP processing framework.

B. Detection Statistic Generation Network Structure

The proposed detection statistics generation network employs a dual-branch input structure, receiving the clutter space-time spectrum and the complex signal of the target cell, and enhances detection accuracy through a hierarchical feature fusion strategy across three core modules. The specific structure is as follows:

Space-Time Spectrum Encoder: This module receives the input clutter space-time spectrum and employs convolutional kernels of sizes 3×3 and 5×5 in parallel to extract multi-scale features. After feature fusion, an attention mechanism

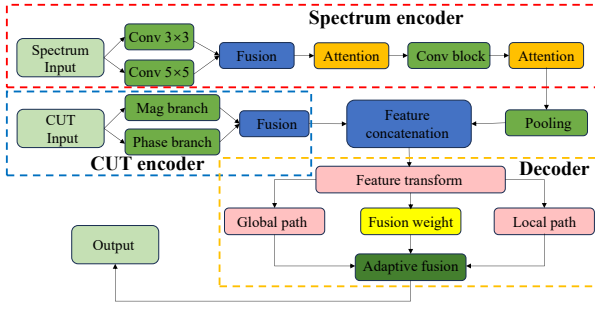


Fig. 3. Nonlinear detection statistics network structure.

is incorporated to enhance the weight of critical regions, ultimately outputting a deep space-time feature map.

Complex CUT Encoder: This component separately extracts the amplitude and phase features from the input complex signal using multiple layers of one dimensional convolution followed by global average pooling. Cross path feature fusion is applied to retain and integrate the joint information of amplitude and phase.

Decoder: This module concatenates the features output by the two encoders, forming a unified multimodal feature representation. Further, through feature transformation and adaptive fusion methods, it effectively combines global context with local details. The spatial resolution is incrementally restored through three layers of transposed convolution, and finally, a dynamic calibration is performed on the output features using an attention gated mechanism, generating the detection statistic corresponding to the target range cell.

C. Dataset and Training

In this work, the network is trained on a simulation generated dataset that incorporates various factors, including terrain and landform variations, range cells, yaw angles, array amplitude phase errors, and clutter fluctuations, to effectively simulate radar echo characteristics in practical nonhomogeneous clutter environments. To improve training efficiency, a two stage training strategy is adopted. First, the nonlinear pulse compression network, nonlinear clutter space-time spectrum estimation network, and nonlinear detection statistic generation network are independently pre-trained using three separate datasets. Subsequently, using a unified dataset, the parameters of the first two networks are frozen, and only the nonlinear detection statistic generation network is fine tuned. Each training dataset contains 10,000 samples. During the testing phase, a complete dataset of radar echoes and their corresponding detection statistics is generated. The end-to-end network outputs are then directly evaluated and analyzed in subsequent simulation experiments.

To ensure that the network can accurately reconstruct both the overall amplitude structure and the critical target features, we have designed a loss function composed of three components: the global mean squared error term, the peak preservation term, and the constraint term for the secondary peak regions. The overall loss is defined as

$$\mathcal{L} = \alpha \mathcal{L}_{\text{MSE}} + \beta \mathcal{L}_{\text{peak}} + \gamma \mathcal{L}_{\text{region}}. \quad (3)$$

where, α , β and γ denote the weighting coefficients. The global term

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N (z_i - \hat{z}_i)^2 \quad (4)$$

ensures overall reconstruction fidelity. To preserve the main-lobe peak, which is critical for target detection, the maximum values of the target and prediction are extracted as $p_{\text{pred}} = \max(\hat{z})$ and $p_{\text{true}} = \max(z)$, yielding the peak loss

$$\mathcal{L}_{\text{peak}} = (p_{\text{pred}} - p_{\text{true}})^2. \quad (5)$$

To constrain the sidelobe and secondary peak structure, an additional loss is applied to elements within a specified amplitude range. The central region, excluded based on prior knowledge of the clutter ridge location, is masked to avoid bias. Denoting the effective index set by $\mathcal{M}$, the regional loss is

$$\mathcal{L}_{\text{region}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} (z_i - \hat{z}_i)^2. \quad (6)$$

IV. SIMULATED RESULTS

A. Experimental Setup

This section evaluates the proposed nonlinear end-to-end STAP network through simulations and compares it with optimal (OPT) STAP [3], sparse Bayesian learning (SBL) based STAP [16], and CNN based space-time spectrum estimation STAP [26]. The OPT STAP method serves as the performance upper bound.

The dataset of proposed method is randomly divided into training and testing subsets with an 8:2 ratio. All simulations are conducted on a workstation equipped with an Intel(R) Core(TM) i7-11700K@3.60 GHz CPU and an NVIDIA GeForce RTX 3070 GPU, using PyTorch [28] and MATLAB for implementation. The network is trained for 300 epochs with an initial learning rate of 1×10^{-3} , which is updated using a cosine annealing schedule. A dropout rate of 0.15 and a weight decay of 5×10^{-4} are employed to improve generalization. The key radar system parameters used in the simulations are summarized in Table II.

B. Spectrum Reconstruction Results

The performance of the proposed method is assessed by comparing the clutter suppressed range-Doppler spectra with those obtained by representative benchmark methods. As illustrated in Fig. 4, the proposed approach exhibits consistently superior clutter suppression. Specifically, it achieves a noticeably narrower mainlobe compared with SBL STAP and CNN STAP, while the residual clutter power in the sidelobe region is significantly lower and closest to that of OPT STAP. The above results indicate that, due to the end-to-end nonlinear processing structure employed by the proposed method, it avoids the model mismatch loss caused by inconsistencies between the nonlinear processing preset model and subsequent linear processing, as seen in the comparison methods, thus achieving stronger clutter suppression.

TABLE I
COMPUTATIONAL COMPLEXITY AND RUNTIME COMPARISON

Method	Complexity	Runtime
SBL STAP	$O((NK N_S N_D + (NK)^3 + 3(NK)^3 N_S N_D + 2NK(N_S N_D)^2)I_{\text{SBL}})$	130.42 s
CNN STAP	$O(28777 N_S N_D + (NK)^2 N_S N_D)$	1.63 s
Proposed	$O(71677 N_S N_D)$	2.30 s

TABLE II
RADAR SYSTEM PARAMETERS

Parameter	Value
PRF	2600 Hz
Platform velocity	150 m/s
Platform height	10 km
Number of antenna elements	8
Pulses per CPI	32
Wavelength	0.2308 m
Bandwidth	2.0 MHz

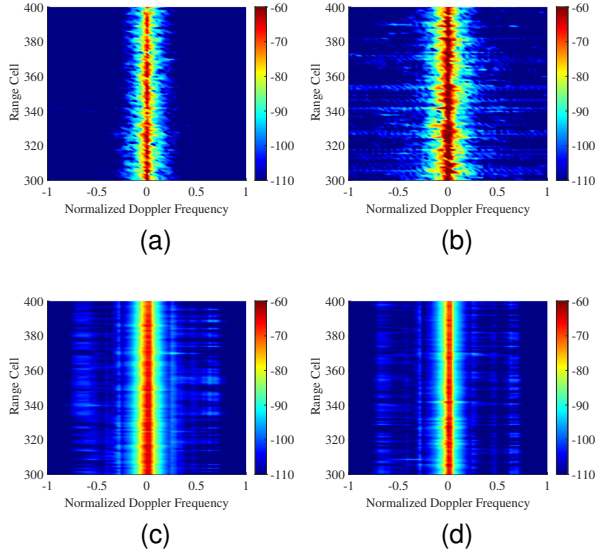


Fig. 4. Comparison of range Doppler spectra of various methods. (a) OPT STAP. (b) SBL STAP. (c) CNN STAP. (d) Proposed method.

C. Target Detection Results

In this test, range slices are used to compare the target detection performance of different methods after clutter suppression. The test data includes target 1 placed in the mainlobe region (range cell 370) and target 2 placed in the sidelobe region (range cell 340), with the average output signal-to-clutter-plus-noise ratio (SCNR) used as the comprehensive performance metric.

As shown in Fig. 5(a), for the target located in the mainlobe clutter region, the proposed method effectively separates it from the background clutter, and the residual clutter power level after processing is comparable to that of the OPT STAP method. In terms of average SCNR, the proposed method outperforms SBL STAP and CNN STAP by 5.59 dB and 7.94 dB, respectively. Furthermore, as illustrated in Fig. 5(b), in the sidelobe target detection scenario, the average SCNR

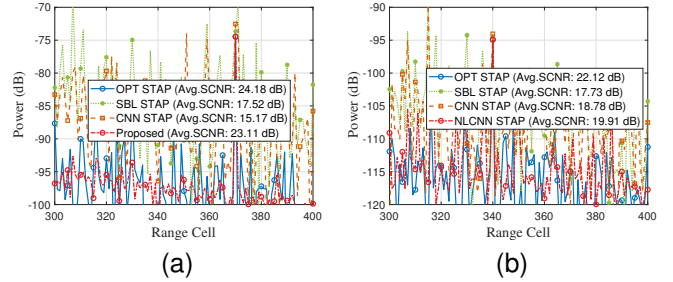


Fig. 5. Comparison of clutter suppression performance. (a) Target 1. (b) Target 2.

of the proposed method is 2.18 dB and 1.13 dB higher than that of SBL STAP and CNN STAP, respectively. These experimental results demonstrate that the proposed method exhibits significant performance advantages and delivers excellent and robust target detection performance compared to the benchmark methods.

D. Computational Complexity

This subsection compares the computational complexity and online runtime of the proposed method with representative nonlinear STAP algorithms, as summarized in Table I. In these expressions, I_{SBL} denotes the number of SBL iterations, which is set to 100 following common practice, and N_S and N_D represent the spatial and temporal discretization grid sizes used in spectrum estimation. The results show that the proposed method achieves a complexity and runtime comparable to CNN STAP, while substantially outperforming the more computationally intensive SBL STAP.

V. CONCLUSION

This letter proposes an end-to-end deep learning STAP method that integrates three core subnetworks for nonlinear pulse compression, nonlinear space-time spectrum estimation, and nonlinear clutter statistic generation. Simulation results indicate that the proposed method outperforms conventional nonlinear STAP methods in clutter suppression and target detection, while considerably lowering the computational burden relative to sparse recovery techniques. The proposed method is well suited for practical nonhomogeneous clutter environments where training samples are limited. Future work will focus on validating its effectiveness using real measured radar data.

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